

Article

Soil Solution Viscosity Reduces CO₂ Emissions in Tropical Soils: Implications for Climate Change Mitigation

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Abstract

Soil CO₂ emissions, driven primarily by microbial respiration, represent a major component of terrestrial carbon flux and play a crucial role in global climate change. Although several soil physicochemical factors regulating microbial activity are well known, the role of soil solution viscosity remains largely unexplored. This study evaluated how polyethylene glycol (PEG6000)-induced increases in soil solution viscosity affect microbial activity-derived CO₂ emissions in a Rhodic Ferralsol (eutric). Three concentrations of PEG6000 (50, 75, and 100 g L⁻¹), corresponding to viscosities of 1.93, 2.76, and 3.88 cP, respectively, were compared to a water-based control (1.11 cP). Soil CO₂ emissions, soil O₂ capture, temperature, and water content were measured over a 60-day period using standard methods. Results showed significant reductions in cumulative CO₂ emissions of 20%, 25%, and 12% for PEG6000 treatments, respectively, compared to the control. Decreased O₂ capture at viscosities of 1.93 and 2.76 cP (50 and 75 g L⁻¹, respectively) indicated reduced microbial activity. These findings reveal a previously underappreciated biophysical mechanism regulating soil carbon emissions. Understanding and managing soil solution viscosity could offer a novel strategy to mitigate CO₂ emissions in tropical soils, thus contributing to climate change mitigation and sustainable soil management, particularly in highly weathered tropical ecosystems.

Keywords: biophysical control; CO₂ emissions; microbial activity; soil carbon protection; soil respiration; soil solution; sustainable soil management



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1. Introduction

Global carbon dioxide (CO₂) emissions reached approximately 10.9 Pg-C yr⁻¹ during the last decade, representing a 144% increase compared to pre-industrial levels. Alongside methane (CH₄) and nitrous oxide (N₂O), which emitted between 335 and 383 Tg CH₄ yr⁻¹ and 4.2–11.4 Tg N₂O yr⁻¹ between 2010 and 2019, these gases constitute the main contributors to anthropogenic climate change [1]. The predominant sources of these

emissions include fossil fuel combustion and activities related to land use change and management, such as agriculture, forestry, and other land uses (AFOLU) [2]. Current atmospheric CO₂ concentrations are ~427 ppm [3], whereas projections indicate that by 2100, CO₂ levels could range from 393 ppm in the low-emission scenario (SSP1-1.9) to 1135 ppm in the very high-emission scenario (SSP5-8.5) [4].

Between 2010 and 2019, AFOLU contributed between 13 and 21% of global greenhouse gas emissions, equivalent to 5.9 Gt CO₂, 157 Mt CH₄, and 6.6 Mt N₂O per year [2]. In Brazil, emissions derived from agricultural products and land use changes accounted for 25.8% and 38.0% of the national total of CO₂-eq in 2020, respectively [5]. To mitigate these emissions, initiatives such as the global RECSOIL program have promoted sustainable soil management to increase soil organic carbon (SOC) [6]. Brazil has surpassed its national emission reduction targets, committing to a 37% reduction by 2025 and a 50% reduction by 2030, relative to 2005 levels [5,7].

AFOLU is not only a source of emissions but also acts as a carbon sink, with significant potential to mitigate climate change through carbon capture and storage. Agriculture is a key sector for large-scale short-term CO₂ removal, primarily through SOC management [2,8,9], which constitutes the largest terrestrial carbon reservoir, with approximately 677 Pg in the top 30 cm of soil and a global sequestration capacity estimated at 2.45 Pg-C yr⁻¹ [10].

To design effective strategies for mitigating soil carbon loss, it is crucial to understand microbial respiration, which is responsible for a large proportion of organic matter (OM) decomposition and, consequently, CO₂ release [11]. Microorganisms access OM through inter- and intra-aggregate soil pores, depending on their motility in the soil solution [12–14]. In tropical soils, favorable temperature and moisture conditions increase OM mineralization due to intensified microbial activity [15–17]. However, studies are still needed to elucidate the mechanisms regulating this motility and mineralization to optimize carbon sequestration and reduce emissions.

The soil solution, which harbors most microorganisms, presents physical properties such as viscosity that influence microbial flow and movement [14,18]. In other liquid media, such as marine waters or culture media, viscosity has been shown to affect microbial motility rates [19–24]. Recently, it has been hypothesized that soil solution viscosity could regulate microbial CO₂ emissions by affecting microbial diffusion, although this hypothesis lacks empirical evidence under real conditions [25].

Some biopolymers, such as gelatin, sodium alginate, xanthan gum, and polyacrylamide, have been used as viscosity-modifying agents to reduce evapotranspiration; however, they have shown limited soil infiltration [26]. In contrast, polyethylene glycol 6000 (PEG6000), which is water-soluble and low in toxicity, can infiltrate without affecting plant health [27,28] because its molecules are too large to penetrate microbial or plant cells [29]. This makes it an ideal candidate for studying the effects of increased soil solution viscosity on microbial respiration.

This study hypothesizes that soil solution viscosity regulates microbial CO₂ emissions. The objective was to evaluate the effect of different viscosities induced by PEG6000 on soil CO₂ emissions.

2. Materials and Methods

2.1. Study Area

This study was conducted from September to November 2021 at the Agroclimatology Experimental Station of the School of Agricultural and Veterinary Sciences of São Paulo State University “Júlio de Mesquita Filho” (UNESP) in Jaboticabal, São Paulo, Brazil (21°14' S, 48°17' W; 615 m above sea level). The regional climate is classified as

B1rB'4a' [30,31], mesothermally humid, with an average annual precipitation of 1430 mm and an average annual temperature of 22.2 °C [32]. The local soil was classified as Rhodic Ferralsol (eutric) according to the FAO-WRB classification system [33].

2.2. Experimental Design and Treatments

Four treatments were evaluated, consisting of polyethylene glycol 6000 (PEG6000) solutions at concentrations of 0, 50, 75, and 100 g PEG6000 L⁻¹, labeled as C₀, C₅₀, C₇₅, and C₁₀₀, respectively, to increase the viscosity of the solution applied to the soil.

The treatments were applied once at the beginning of the experiment by uniformly distributing a 2 mm layer of PEG6000 solution over the surface of each lysimeter (1 m²) using a manual watering can. This methodology aimed to ensure homogeneous coverage of the treated area, promoting the infiltration of the solution into the soil profile and facilitating contact with the active microbial community at a depth of 0.10 m. The use of lysimeters prevented interference between treatments and concentrated their effects within a restricted area (Figure 1).

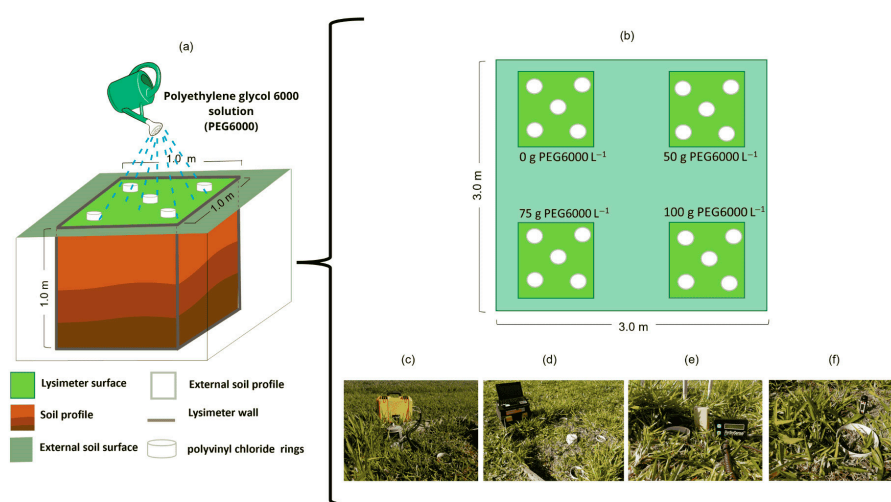


Figure 1. Schematic representation of the experimental arrangement of the treatments and sampling points. (a) Dimensions of the lysimeter and its components. (b) Spatial arrangement of lysimeters. Measurement of variables: (c) Soil CO₂ emissions. (d) Soil O₂ capture. (e) Soil water content. (f) Soil temperature.

Each adjacent lysimeter had an area and depth of 1 m² and 1 m, respectively. The total area of the experimental site, including the four lysimeters and the space between them, was 9 m² (Figure 1). The lysimeters consisted of cement boxes built in the ground and filled with local soil, including native vegetation cover (grass). Built in 1985, these structures have maintained stable physical properties over time and accurately represent the soil conditions surrounding them.

The viscosities of the applied solutions were estimated by using an empirical relationship for PEG6000 viscosity at 25 °C, as reported by [34]. A nonlinear regression model was fitted to the data from that study to relate PEG6000 concentration with viscosity, and this equation was then used to model the viscosities at 50, 75, and 100 g PEG6000 L⁻¹ concentrations.

To measure soil CO₂ emissions (FCO₂) and soil O₂ capture (FO₂), five polyvinyl chloride (PVC) rings were inserted into the surface of each lysimeter, evenly distributed over the area treated with PEG6000. Each ring, measuring 0.10 m in diameter and 0.04 m in height, served as a coupling interface for the gas measurement chambers, forming five sampling points per treatment group for gas measurements. The same sampling points were used to assess the soil temperature (Ts) and soil water content (SWC) (Figure 1). The

variables were measured concurrently on 25 sampling days distributed over a 60-day period between 7:00 and 9:00 a.m., and the sampling days were reported as Julian days. Climatic data (global solar radiation, air temperature, atmospheric pressure, and precipitation) for the study period were obtained from an agroclimatology station located at the same site (Figure 2).

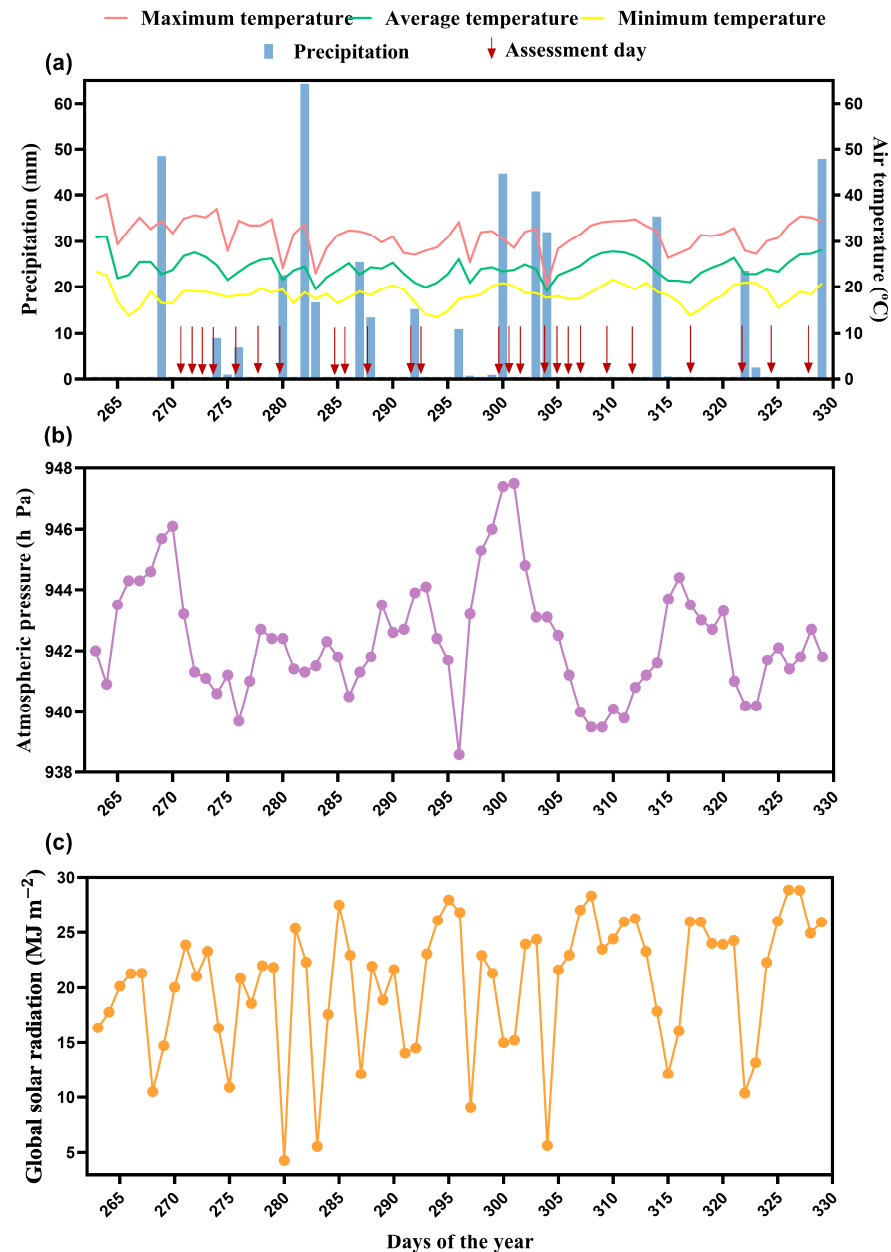


Figure 2. Meteorological data during the study period. (a) Bar graph indicates the precipitation (mm); line graphs show the air temperature (°C) (from top to bottom: maximum, average, and minimum); red arrows indicate the assessment days. (b) Atmospheric pressure (hPa). (c) Global solar radiation (MJ m⁻²).

2.3. CO₂ Emission, Soil Temperature, and Soil Water Content

Soil CO₂ emissions (FCO₂) were measured at each sampling point using an automated portable system (LI-8100, LI-COR, Lincoln, NE, USA), which quantifies changes in soil CO₂ concentration within a closed chamber connected to PVC collars inserted into the soil using an infrared gas analyzer (IRGA). The total soil CO₂ emissions over the 60-day study period were calculated by integrating FCO₂ over time.

Soil temperature (T_s) was recorded in the top 0–0.10 m soil layer using a digital thermometer. The soil water content (SWC) at the same depth was determined using portable time-domain reflectometry (TDR-Campbell® system by Hydrosense TM, Campbell Scientific, Garbutt, QLD, Australia). The system was equipped with a probe consisting of two rods, each 12 cm in length, which were inserted vertically into the soil perpendicular to the surface.

2.4. O_2 Capture

The oxygen concentration was measured using a fluorescence quenching sensor with ultraviolet (UV) light at 0–25% CO_2 , using a portable UV flux sensor (CO_2 Meter, Inc., Ormond Beach, FL, USA). The UV sensor was connected to a computer, enabling real-time data acquisition through the GasLab® software v2.3.1.4, which was also used to configure and calibrate the device. Soil O_2 capture was calculated using the methodologies described by [35,36], by applying the equations presented below.

The soil O_2 diffusion rate (dO_2/dt) was calculated using the linear interpolation of the oxygen concentration inside the chamber over time Equation (1):

$$FO_2(t) = \frac{dO_2}{dtA} \quad (1)$$

where $FO_2(t)$ is the O_2 concentration calculated over time, dO_2 is the change in concentration with respect to time (dt), and A is the collar-surface area [37,38]. The sensor was mounted on a 0.055 m high PVC cap placed on a 0.03 m high PVC collar, forming a 0.085 m high chamber with a volume of 0.00066 m^3 (V) and a soil contact area of 0.008 m^2 (A).

The volume measured using the sensor (ppm) was converted to mol of O_2 using the ideal gas equation (Equation (2)):

$$P(\Delta V) = (\Delta n) RT \quad (2)$$

where P is the mean atmospheric pressure (Pa) over the study period (Figure 2b); ΔV is the variation of captured O_2 (ppm), times the chamber volume, times 1×10^{-6} ($m^{-3} s^{-1}$); R is the constant for two perfect gases ($8.31 J mol^{-1} K^{-1}$); and T is at air temperature (K). The results are expressed in terms of variation of O_2 (Δn) ($mol s^{-1}$).

Soil oxygen capture (FO_2) in $mg m^{-2} s^{-1}$ was determined by linear interpolation of concentration data as a function of time, considering variables such as atmospheric pressure, temperature, and volume of gas retained in the chamber [39,40], as shown in Equation (3):

$$FO_2 = \frac{dO_2}{dt} \frac{10^{-6} PM}{RT} H \quad (3)$$

where FO_2 is the oxygen capture ($mg m^{-2} s^{-1}$); dO_2/dt is the amount of O_2 (ppm) measured at time t (s); P is the mean atmospheric pressure (Pa) over the study period; M is the molar mass of O_2 ($g m^{-3}$); R is the universal gas constant ($8.31 J mol^{-1} K^{-1}$); T is the absolute temperature (K); and $H = V/A$ is the ratio of the chamber volume (V) to the area (A) of the chamber placed on the ground surface.

Similar to CO_2 emissions, total soil O_2 capture up to day 60 of this study was calculated by integrating FO_2 over time [41].

2.5. Data Processing and Analysis

Descriptive statistics (mean, standard deviation, median, interquartile range [IQR], minimum, maximum, standard error of the mean, and coefficient of variation) were calculated for all variables (Table A1). Normality (Shapiro–Wilk test) and homoscedasticity (Bartlett's test) were evaluated at a 5% significance level to verify ANOVA assumptions.

Because the data did not meet these assumptions, cumulative soil CO₂ emissions and O₂ capture were log-transformed, soil temperature (Ts) was transformed using the lambda method, and soil water content (SWC) was square-root transformed. However, all results are presented in the original scale in the figures and tables.

Temporal variability was analyzed using repeated measures ANOVA at a 5% significance level, complemented by regression analyses of FCO₂, FO₂, Ts, and SWC. All statistical analyses were performed in the R environment [42].

3. Results

3.1. Viscosity

The nonlinear regression model shown in (Figure 3a) was generated in the present study based on the tabulated data reported by [34], which describes the relationship between viscosity at 25 °C and PEG6000 concentration. These data were extracted exclusively for comparative and analytical purposes only. The model showed a strong positive correlation, with viscosity increasing as a function of PEG6000 concentration, and exhibited a high degree of fit ($R^2 = 0.99$).

Based on this regression, the viscosities corresponding to the PEG6000 solutions used in the experimental treatments were estimated (Figure 3b). The modeled values were 1.11 cP for the control (C₀, tap water), 1.93 cP for C₅₀, 2.76 cP for C₇₅, and 3.88 cP for C₁₀₀. These results confirmed the concentration-dependent increase in viscosity, supported by a coefficient of determination of $R^2 = 1.00$.

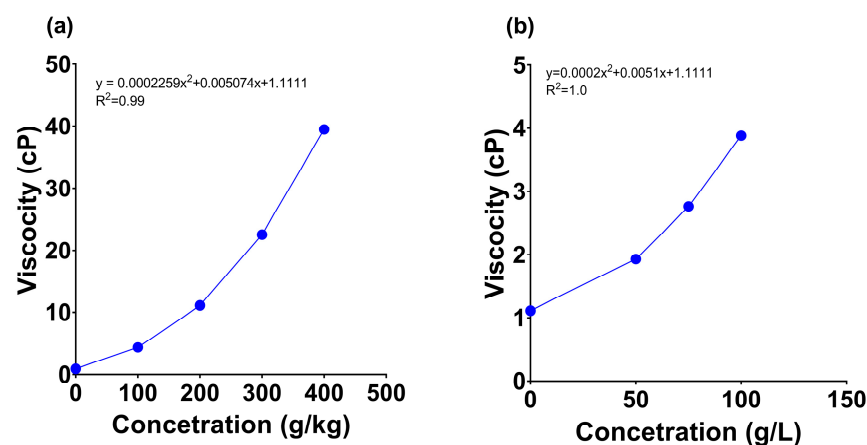


Figure 3. PEG6000 viscosity-concentration relationship: (a) Nonlinear regression model generated in the present study based on tabulated data reported by [34], describing the relationship between viscosity at 25 °C and PEG6000 concentration. Data from [34] were extracted solely for comparative and analytical purposes in this study. (b) Estimated viscosities of the PEG6000 solutions applied in the experimental treatments of the present study, calculated from the regression shown in (a).

3.2. Temporal Variation of Soil CO₂ Emission, Soil O₂ Capture, Soil Temperature, and Soil Water Content

The analysis of variance (ANOVA) for soil CO₂ emissions (FCO₂) revealed a significant effect of treatment ($F = 5.04$; $p < 0.01$), temporal variation (days) ($F = 51.84$; $p < 0.01$), and the interaction between these factors ($F = 2.77$; $p < 0.01$). Temporal fluctuations in FCO₂ were associated with precipitation events. Five distinct increases in FCO₂ were recorded throughout the monitoring period, each occurring immediately after rainfall. The first increase was observed on day 275, following a 10 mm precipitation event, with FCO₂ values 47–85% higher across all treatments compared to day 273. The highest precipitation-induced peak occurred on day 301 after 44.8 mm of rainfall, with the highest FCO₂ observed in C₀

($6.86 \pm 0.23 \mu\text{mol m}^{-2} \text{s}^{-1}$) and the lowest in C₅₀ ($4.80 \pm 0.63 \mu\text{mol m}^{-2} \text{s}^{-1}$), representing a 30% reduction in C₅₀ relative to C₀ on that day (Figure 4a).

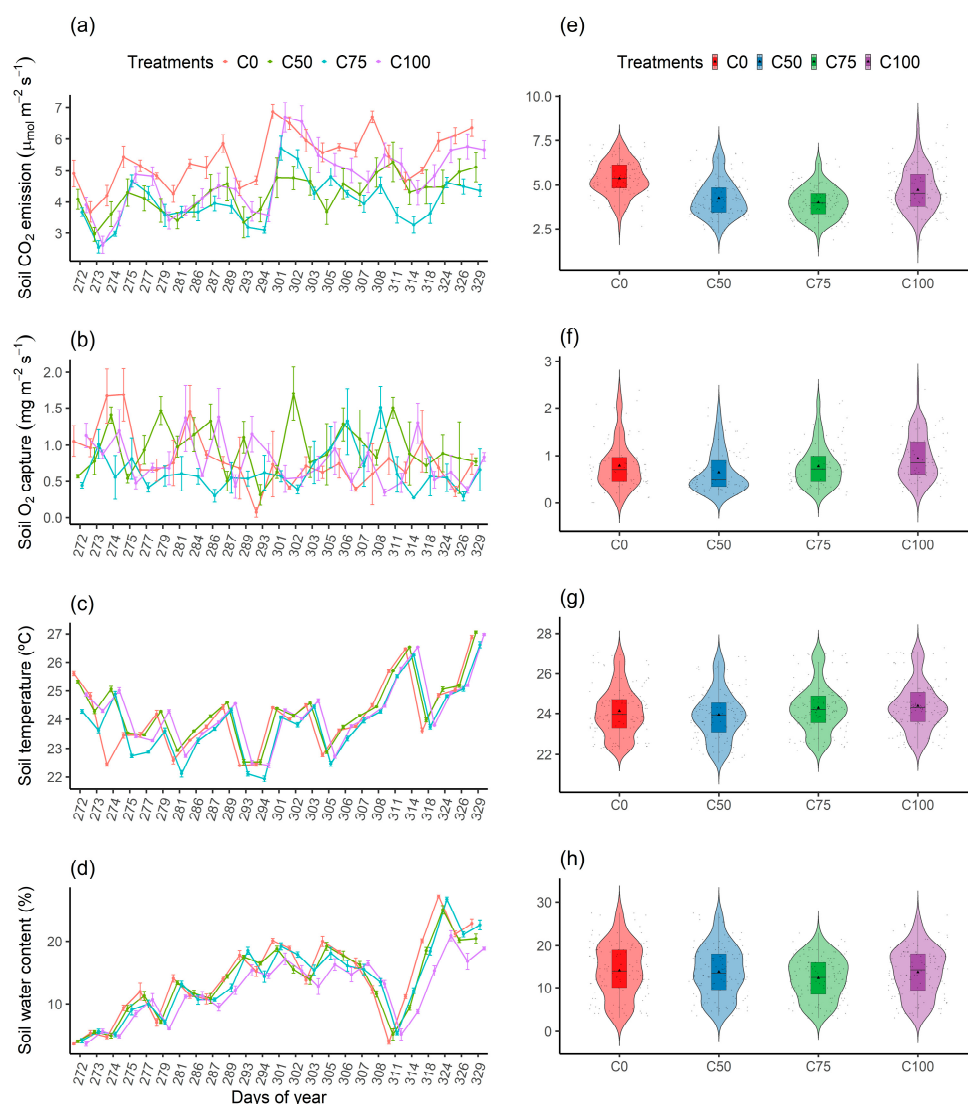


Figure 4. Temporal variability by treatment of the studied variables (daily mean $\pm$ mean standard error): soil CO₂ emission (a), soil O₂ capture (b), soil temperature (c), and soil water content (d). Quadratic regression analysis of means by treatment of the studied: soil CO₂ emission (e), soil O₂ capture (f), soil temperature (g), and soil water content (h). The upper and lower box lines represent the interquartile range (25–75 %); the black line indicates the median and the black triangle indicates the mean; $^{\circ}$ indicates individual variation, and the width of the figures represents the distribution of the data (wider sections represent a greater number of data). The treatments are doses of 0, 50, 75, and 100 g of polyethylene glycol 6000 per liter of water (C₀, C₅₀, C₇₅, and C₁₀₀, respectively).

The lowest FCO₂ values were recorded on day 273, following three days without precipitation. Among the treatments, C₇₅ exhibited the lowest emission ($2.60 \pm 0.20 \mu\text{mol m}^{-2} \text{s}^{-1}$), which was also 30% lower than that of C₀ ($3.69 \pm 0.33 \mu\text{mol m}^{-2} \text{s}^{-1}$) on the same day. Overall, the highest FCO₂ values were observed in the C₀ treatment ($p < 0.05$) on 20 out of the 25 evaluation days, with the most pronounced differences occurring during the first nine days of monitoring ($p < 0.05$). In contrast, the lowest emissions were consistently observed in the C₇₅ treatment (Figure 4a).

Soil O₂ capture (FO₂) differed significantly among the treatments ($F = 2.96$; $p < 0.1$), over time ($F = 2.35$; $p < 0.01$), and in the interaction between these factors ($F = 2.36$; $p < 0.01$).

The highest FO_2 value was observed in treatment C_0 on day 275 ($1.70 \pm 0.36 \text{ mg m}^{-2} \text{ s}^{-1}$), whereas the lowest was recorded in the same treatment on day 293 ($0.07 \pm 0.07 \text{ mg m}^{-2} \text{ s}^{-1}$), showing a decreasing trend. From day 297, the FO_2 in C_0 remained stable until the end of this study (Figure 4b).

The other treatments exhibited greater variability in FO_2 during this study. In C_{50} , FO_2 fluctuated between $1.50 \pm 0.30 \text{ mg m}^{-2} \text{ s}^{-1}$ on day 308 and $0.30 \pm 0.00 \text{ mg m}^{-2} \text{ s}^{-1}$ on day 314. In C_{75} , the maximum and minimum FO_2 values were 1.37 ± 0.40 and $0.34 \pm 0.05 \text{ mg m}^{-2} \text{ s}^{-1}$, recorded on days 286 and 308, respectively. Treatment C_{100} showed an increasing trend in FO_2 , rising from $0.32 \pm 0.14 \text{ mg m}^{-2} \text{ s}^{-1}$ on day 293 to $1.70 \pm 0.37 \text{ mg m}^{-2} \text{ s}^{-1}$ on day 392 (Figure 4b).

Soil temperature (T_s) varied significantly over time ($F = 1854.98$; $p < 0.01$), among treatments ($F = 22.69$; $p < 0.01$), and in the interaction between these factors ($F = 34.31$; $p < 0.01$). The lowest T_s was recorded in the C_{50} treatment on day 294 ($21.94 \pm 0.10 \text{ }^\circ\text{C}$; $p < 0.05$). In contrast, the highest T_s was observed in the C_{100} treatment on day 329 ($27.1 \pm 0.05 \text{ }^\circ\text{C}$). On that day, the maximum T_s values across treatments were $26.89 \pm 0.07 \text{ }^\circ\text{C}$ (C_0), $26.57 \pm 0.12 \text{ }^\circ\text{C}$ (C_{50}), and $26.98 \pm 0.05 \text{ }^\circ\text{C}$ (C_{75}), with C_{50} showing significantly lower values ($p < 0.05$) (Figure 4c).

A decreasing trend in T_s was observed from the beginning of this study until day 294, after which the values remained relatively stable. This decrease coincided with the beginning of rainfall events and elevated SWC, suggesting a potential link that may be related to the onset of precipitation events and the associated increase in soil moisture.

Soil water content (SWC) varied significantly over time ($F = 448.29$; $p < 0.01$), among treatments ($F = 4.49$; $p < 0.01$), and in the interaction between these factors ($F = 2.80$; $p < 0.01$). The lowest SWC values were recorded on day 272 across all treatments: $3.8 \pm 0.12\%$ (C_0), $4.3 \pm 0.34\%$ (C_{50}), $3.8 \pm 0.34\%$ (C_{75}), and $4.2 \pm 0.20\%$ (C_{100}) (Figure 4d). The highest SWC was observed on day 324 in all treatments, although C_{75} exhibited the lowest moisture content on that day: $27.00 \pm 0.27\%$ (C_0), $26.50 \pm 0.35\%$ (C_{50}), $20.90 \pm 0.81\%$ (C_{75}), and $24.87 \pm 0.64\%$ (C_{100}).

Between days 305 and 311, a marked decline in SWC was recorded owing to the absence of precipitation and elevated soil temperatures. During this period, the moisture content decreased by approximately 80%, 69%, 68%, and 72% in C_0 , C_{50} , C_{75} , and C_{100} , respectively.

3.3. Quadratic Regression Analysis of Soil CO_2 Emission, Soil O_2 Capture, Soil Temperature, and Soil Water Content

Quadratic regression analyses were performed to evaluate the effects of PEG6000 concentrations on soil CO_2 emissions (FCO_2), soil O_2 capture (FO_2), soil temperature (T_s), and soil water content (SWC) (Table 1). The analysis of FCO_2 revealed a significant influence of the treatments, with a high coefficient of determination ($R^2 = 0.95$). The emissions decreased at C_{50} ($4.26 \pm 0.09 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$), C_{75} ($4.00 \pm 0.08 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$), and C_{100} ($4.72 \pm 0.11 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$) compared to the control C_0 ($5.38 \pm 0.08 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$), corresponding to reductions of 20.8%, 25.6%, and 12.3%, respectively (Figure 4e). The model predicted a minimum FCO_2 of $4.22 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$ at $54.12 \text{ g PEG6000 L}^{-1}$.

For FO_2 , the quadratic model indicated a decrease at C_{50} ($0.65 \pm 0.05 \text{ mg m}^{-2} \text{ s}^{-1}$), representing a 17.72% reduction relative to C_0 ($0.79 \pm 0.06 \text{ mg m}^{-2} \text{ s}^{-1}$), while C_{75} ($0.77 \pm 0.05 \text{ mg m}^{-2} \text{ s}^{-1}$) showed a similar capture to C_0 , and C_{100} ($0.95 \pm 0.06 \text{ mg m}^{-2} \text{ s}^{-1}$) exhibited a 20% increase. This model presented a strong fit, with $R^2 = 0.99$ (Figure 4f). In contrast, soil temperature showed no significant variation across the treatments (C_0 : $24.1 \pm 0.11 \text{ }^\circ\text{C}$; C_{50} : $23.9 \pm 0.11 \text{ }^\circ\text{C}$; C_{75} : $24.2 \pm 0.10 \text{ }^\circ\text{C}$; C_{100} : $24.3 \pm 0.10 \text{ }^\circ\text{C}$) (Figure 4g), with a moderate coefficient of determination ($R^2 = 0.79$) and non-significant parameters for the regression equation. Finally, the analysis of SWC resulted in a low fit ($R^2 = 0.44$) and

parameters that were not normally distributed; nevertheless, a notably lower SWC was observed in C₇₅ ($12.20 \pm 0.43\%$) than in C₀ ($14.10 \pm 0.56\%$) (Figure 4h), possibly reflecting treatment-induced changes in soil solution viscosity.

Table 1. Quadratic regression analysis for soil CO₂ emissions, soil O₂ capture, soil temperature, soil water content, total soil CO₂ emissions, and total soil O₂ capture.

Variable	Variable Quadratic Regression = $b_0 + b_1 C + b_2 C^2$			RMSE	Minimum	
	b_0	b_1	b_2		X	Y
FCO ₂ ⁽¹⁾	5.3958 ± 0.2217 ($p = 0.026$)	-0.0433 ± 0.0101 ($p = 0.14$)	0.0004 ± 0.0001 ($p = 0.17$)	0.95	0.2681	54.13
FO ₂ ⁽²⁾	0.7882 ± 0.0190 ($p = 0.015$)	$-0.0067 \pm$ 0.0008 ($p = 0.08$)	0.0001 ± 0.000008 ($p = 0.06$)	0.99	0.0296	33.50
Ts ⁽³⁾	24.0873 ± 0.023 ($p = 0.003$)	-0.0069 ± 0.0061 ($p = 0.46$)	0.0001 ± 0.0001 ($p = 0.36$)	0.79	0.0709	38.33
SWC ⁽⁴⁾	14.2027 ± 1.0725 ($p = 0.05$)	-0.0356 ± 0.0491 ($p = 0.60$)	0.0003 ± 0.0005 ($p = 0.68$)	0.44	0.5746	59.50
TFCO ₂ ⁽⁵⁾	3.199 ± 0.1562 ($p = 0.0311$)	-0.0256 ± 0.0072 ($p = 0.1733$)	0.0002 ± 0.0001 ($p = 0.20$)	0.93	0.1145	64.00
TFO ₂ ⁽⁶⁾	39.225 ± 2.3491 ($p = 0.0381$)	-0.4133 ± 0.1075 ($p = 0.1620$)	0.0049 ± 0.0011 ($p = 0.14$)	0.96	1.1836	42.17

FCO₂: Soil CO₂ emission. FO₂: Soil O₂ capture. St: soil temperature. SWC: Soil water content. TFCO₂: Total soil CO₂ emission. TFO₂: Total soil O₂ capture. ⁽¹⁾ b_0 : $\mu\text{mol m}^{-2} \text{s}^{-1}$. b_1 : $\mu\text{mol m}^{-2} \text{s}^{-1} (\text{g L}^{-1})^{-1}$. b_2 : $\mu\text{mol m}^{-2} \text{s}^{-1} (\text{g L}^{-1})^{-2}$. ⁽²⁾ b_0 : $\text{mg m}^{-2} \text{s}^{-1}$. b_1 : $\text{mg m}^{-2} \text{s}^{-1} (\text{g L}^{-1})^{-1}$. b_2 : $\text{mg m}^{-2} \text{s}^{-1} (\text{g L}^{-1})^{-2}$. ⁽³⁾ b_0 : °C. b_1 : °C $(\text{g L}^{-1})^{-1}$. b_2 : °C $(\text{g L}^{-1})^{-2}$. ⁽⁴⁾ b_0 : Vol%. b_1 : Vol% $(\text{g L}^{-1})^{-1}$. b_2 : Vol% $(\text{g L}^{-1})^{-2}$. ⁽⁵⁾ b_0 : ton ha^{-1} . b_1 : $\text{ton ha}^{-1} (\text{g L}^{-1})^{-1}$. b_2 : $\text{ton ha}^{-1} (\text{g L}^{-1})^{-2}$. ⁽⁶⁾ b_0 : ton ha^{-1} . b_1 : $\text{ton ha}^{-1} (\text{g L}^{-1})^{-1}$. b_2 : $\text{ton ha}^{-1} (\text{g L}^{-1})^{-2}$.

3.4. Quadratic Regression Analysis of Total Soil CO₂ Emission and Total Soil O₂ Capture

The total soil CO₂ emissions averaged 3.185 ± 0.08 , 2.544 ± 0.25 , 2.365 ± 0.009 , and 2.825 ± 0.18 tons CO₂ ha^{−1} for treatments C₀, C₅₀, C₇₅, and C₁₀₀, respectively (Figure 5a). Quadratic regression analysis confirmed the significant effect of the treatments on total emissions, with a coefficient of determination of 0.93 (Table 1). Specifically, treatments C₇₅, C₅₀, and C₁₀₀ reduced soil CO₂ emissions by 25.7%, 20.1%, and 11.3%, respectively, compared with C₀. These findings support the hypothesis that solution viscosity influences cumulative CO₂ emissions.

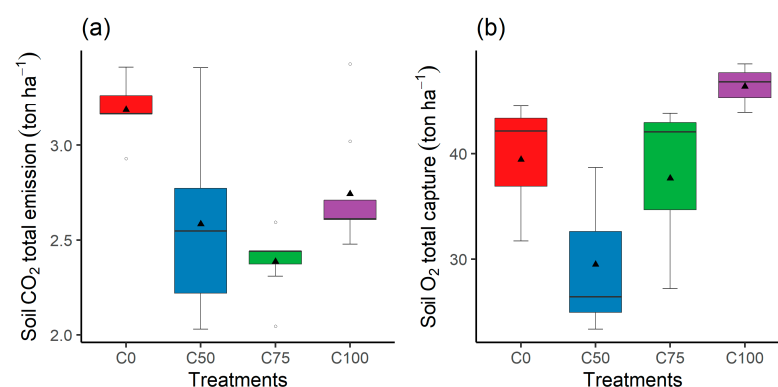


Figure 5. Quadratic regression analysis by treatment. Box plots for total soil CO₂ emission (a) and total soil O₂ capture (b), accumulated during the experiment. The treatments are doses of 0, 50, 75, and 100 g of polyethylene glycol 6000 per liter of water (C₀, C₅₀, C₇₅, and C₁₀₀, respectively). Triangles represent means.

Similarly, the average total soil O₂ capture was 39.45 ± 3.93 , 29.5 ± 4.68 , 37.7 ± 5.25 , and 46.40 ± 1.35 tons ha^{−1} for treatments C₀, C₅₀, C₇₅, and C₁₀₀, respectively (Figure 5b).

Capture in C_{100} was 17.57% higher than that in C_0 , whereas C_{50} and C_{75} exhibited reductions of 25.22% and 4.23%, respectively. Quadratic regression corroborated these trends with a determination coefficient of 0.96 (Table 1).

4. Discussion

In this study, the temporal variability of soil CO_2 emissions (FCO_2) was closely related to changes in soil water content (SWC), which is in agreement with previous reports [16,32,43]. Increases in soil moisture following dry periods likely stimulated microbial respiration, highlighting the crucial role of water in CO_2 diffusion and production [14,44]. This effect was more pronounced at the end of the dry season, with the onset of rainfall, when significant increases in FCO_2 were recorded, consistent with studies conducted in Rhodic Ferralsols and tropical regions [32,35,45,46]. Conversely, after continuous rainfall (days 277 to 281), a reduction in FCO_2 was observed when SWC was higher, likely due to increased water-filled pore space, which limits gas diffusion, as also reported by [32,40].

The direct relationship between soil CO_2 emissions driven by microbial activity and factors such as soil moisture and temperature has been widely explored [11,16]. However, other less-explored physical factors, such as soil solution viscosity, could also influence this process [25]. Our results showed that increasing the viscosity of the applied solution from 1.11 cP in the control (C_0) to 1.93 cP (C_{50}), 2.76 cP (C_{75}), and 3.88 cP (C_{100}) resulted in a reduction in soil CO_2 emissions, with decreases of 20.8%, 25.6%, and 12.3%, respectively. These reductions suggest that higher solution viscosity may act as a limiting factor for microbial respiration, possibly by restricting the movement and access of microorganisms to substrates within the soil matrix. This interpretation is supported by previous studies that have demonstrated a relationship between viscosity and microbial motility in liquid media [20–24].

The observed decrease in oxygen capture (FO_2) throughout the experiment was consistent with the oxygen capture patterns reported in tropical forests [35]. Additionally, the low FO_2 following rainfall events, also reported by [40], may be related to increased soil water content (SWC), which limits gas exchange in soil pores by reducing oxygen diffusion into the soil [36].

Analysis of mean values per treatment showed a decrease in oxygen uptake (FO_2) in C_{50} compared to the control (C_0), suggesting a possible effect of PEG6000. This biopolymer can affect oxygen uptake by hindering its transport due to the increase in medium viscosity, decrease in water potential [47], and increase in osmotic potential, which together reduce water uptake by plant roots [48]. However, this effect was not evident in C_{75} , where the FO_2 values were similar to those of the control. In contrast, C_{100} showed a higher FO_2 than the other treatments, as well as higher CO_2 emissions (FCO_2) than C_{50} and C_{75} .

These results suggest that there was no linear dose-dependent effect on either FCO_2 or FO_2 , possibly because these processes reflect the respiration activity of the microbial community, a process influenced not only by oxygen availability [36] but also by factors such as microbial mobility [25], water content [16], soil physical structure, and access to organic matter [12]. However, the fitted quadratic model for CO_2 (FCO_2) emissions predicted a minimum emission at a concentration of 54.12 g PEG6000 L^{-1} , suggesting the existence of an optimal viscosity point beyond which microbial respiration is maximally restricted, without necessarily indicating a linear relationship between PEG6000 dose and CO_2 emission.

Throughout this study, soil temperature showed low variability between treatments and in average treatment values, which is typical of tropical environments and has been reported in similar Rhodic Ferralsols [16,32,46]. This relative temperature stability likely

favors microbial activity by allowing dormant microorganisms to rapidly activate in response to moisture [49]. Given the low thermal variation observed, the potential effect of temperature on soil CO₂ emissions may have been limited in our study. Nevertheless, temperature can influence the viscosity of the soil solution, which in turn affects microbial motility. According to [25], an increase in temperature from temperate to tropical conditions (0 °C to 30 °C) reduces the viscosity of the soil solution by approximately 55%, from 1.78 to 0.80 mPa·s [11]. Previous studies in marine environments have shown that a 10 °C decrease in temperature increases viscosity and reduces microorganism motility by 4–37% [22].

The variation in soil water content (SWC) during the experiment was clearly influenced by precipitation events, with progressive increases following rainfall and decreases during dry periods, consistent with other studies on Rhodic Ferralsols [32,35]. Although no significant differences were observed in the parameters of the quadratic regression analysis, the average SWC in treatments C₅₀, C₇₅, and C₁₀₀ was lower than that in C₀, which could be related to the effect of PEG6000 in decreasing the osmotic potential of the solutions [29], acting as a stress factor on water availability [50].

We acknowledge the practical limitations associated with the use of PEG6000, particularly its potential adverse effects on plants and microorganisms when applied at high concentrations. These effects are primarily related to its osmotic action, which can induce water stress, as previously reported [50]. In this regard, we suggest that future research evaluate the economic viability of this practice, as well as the use of other thickening agents that are more sustainable, effective, and compatible with plant physiological processes. We also recommend incorporating complementary approaches, such as metagenomics or enzyme activity analysis, to differentiate the physical effects associated with viscosity from the potential chemical or toxicological effects of thickening agents.

5. Conclusions

Our results demonstrate that alterations in soil solution viscosity, experimentally induced using PEG6000, can nonlinearly affect gas diffusion and the dynamics of microbial respiration in the short term. In particular, treatments with intermediate viscosity levels (C₅₀ and C₇₅) reduced both oxygen uptake and CO₂ emissions compared to the control, suggesting a physical limitation on aerobic microbial activity under these conditions. However, in the highest concentration treatment (C₁₀₀), oxygen uptake increased, whereas CO₂ emissions remained lower than those of the control, indicating a more complex response potentially influenced by additional physicochemical or biological factors that require further investigation.

This study represents a pioneering contribution to the understanding of the role of soil solution viscosity in regulating CO₂ emissions, particularly in highly weathered Ferralsols. Despite the practical limitations of PEG6000 for large-scale applications, the results lay the groundwork for developing sustainable strategies that consider viscosity tuning using more suitable, environmentally safe, and economically feasible materials to mitigate climate change.

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Abbreviations

The following abbreviations are used in this manuscript:

CO ₂	carbon dioxide
CH ₄	methane
N ₂ O	nitrous oxide
AFOLU	Agriculture, forestry, and other land uses
RECSOIL	Recarbonization of Global Agricultural Soils
SOC	Soil organic carbon
OM	Organic matter
PEG6000	polyethylene glycol 6000
FCO ₂	Soil CO ₂ emissions ($\mu\text{mol m}^{-2} \text{s}^{-1}$)
FO ₂	Soil O ₂ capture ($\text{mg m}^{-2} \text{s}^{-1}$)
Ts	Soil temperature ($^{\circ}\text{C}$)
SWC	Soil water content (%)
TFCO ₂	Total soil CO ₂ emissions (tons ha^{-1})
TFO ₂	Total soil O ₂ capture (tons ha^{-1})

Appendix A

Table A1. Descriptive statistics of the assessed variables in this experiment.

		N	Mean	SD	Median	IQR	Min	Max	MSE	CV%	Curtose
CO ₂ emission ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	C0	125	5.38	0.96	5.37	1.31	2.65	7.39	± 0.08	17.88	−0.24
	C50	125	4.26	1.07	4.14	1.42	2.24	7.19	± 0.09	25.23	0.13
	C75	125	4.00	0.87	3.95	1.14	1.90	6.47	± 0.08	21.81	0.03
	C100	125	4.72	1.24	4.51	1.87	1.90	8.38	± 0.11	26.21	−0.02
O ₂ capture ($\text{mg m}^{-2} \text{s}^{-1}$)	C0	75	0.79	0.51	0.70	0.51	0.003	2.38	± 0.06	64.30	1.38
	C50	75	0.65	0.43	0.49	0.57	0.15	2.01	± 0.06	66.92	0.91
	C75	75	0.77	0.43	0.70	0.56	0.16	2.15	± 0.05	55.99	1.18
	C100	75	0.95	0.48	0.86	0.71	0.13	2.40	± 0.05	50.79	−0.31
Ts ($^{\circ}\text{C}$)	C0	125	24.10	1.20	24.0	1.40	22.20	27.00	± 0.11	4.98	−0.16
	C50	125	23.90	1.20	23.9	1.40	21.60	26.80	± 0.11	5.02	−0.03
	C75	125	24.20	1.13	24.2	1.30	22.20	27.20	± 0.10	4.67	0.31
	C100	125	24.30	1.11	24.2	1.30	22.30	27.30	± 0.10	4.57	0.44
SWC (v/v) (%)	C0	125	14.10	6.23	14.0	9.00	3.00	27.50	± 0.56	44.18	−0.79
	C50	125	13.70	5.84	13.5	8.50	3.50	27.50	± 0.52	42.63	−0.57
	C75	125	12.20	4.84	12.5	7.50	3.00	23.00	± 0.43	39.67	−0.89
	C100	125	13.60	5.63	14.0	8.50	3.00	27.00	± 0.50	41.39	−0.77

Notes: N is the number of samples; Mean is the mean of the measurements; SD is standard deviation; Median; interquartile range (IQR); Minimum (Min); Maximum (Max); mean standard error (MSE); coefficient of variation (CV); kurtosis of soil CO₂ emission; O₂ capture; soil temperature (Ts); soil water content (SWC) in soil from four lysimeters applied with four doses of polyethylene glycol 6000 (PEG6000) in grams per liter of water; control (C0): 50 g L^{−1} (C50), 75 g L^{−1} (C75), and 100 g L^{−1} (C100).

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